[METRIC]
A-A-59532/3A
March 28, 2006
SUPERSEDING
A-A-59532/3
January 14, 2000

COMMERCIAL ITEM DESCRIPTION

FUSE, INCLOSED LINK, SURFACE MOUNT (SM), 125 VOLTS (V) AC/DC, QUICK ACTING, HIGH BREAKING CAPACITY

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring the fuses described herein shall consist of this document and the issue in effect of CID A-A-59532.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This commercial item description (CID) uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59532	/03	-	001
CID number	Applicable CID specification sheet (two digits)		Dash number

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Fuses supplied to this CID shall be as specified herein (see figure 1).

Electrical specifications.

Ampere rating. The ampere rating shall be as specified in table I.

Voltage rating. The voltage rating shall be 125 V ac/dc.

Time current characteristics. The time current characteristics shall be as indicated in table II.

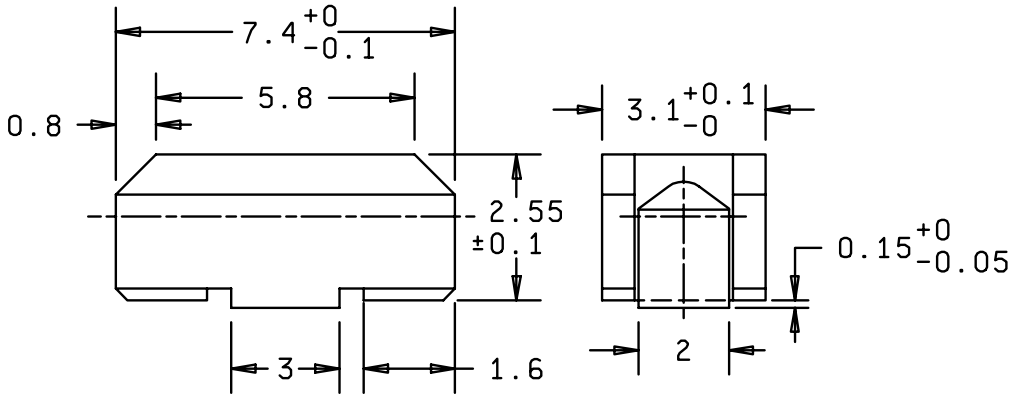
Environmental specifications. Fuses supplied to this CID shall be subject to the following tests and there shall be no electrical or mechanical damage to the fuse.

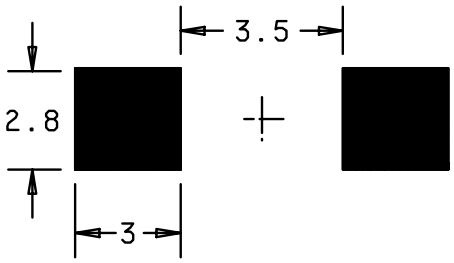
Maximum storage temperature. The maximum storage temperature shall be 40°C (104°F) at 70 percent relative humidity.

Ambient temperature maximum. The ambient temperature shall be -40°C (-40°F) to +85°C (185°F).

Vibration resistance. Fuses shall meet vibration requirements in accordance with IEC 60068-2-6, test Fc. (10-2000 Hz, cross-over frequency 60 Hz, amplitude 0.75 mm, resp. acceleration 100 m/s^2 (10g)).

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SOLDER PADS

mm	Inches	mm	Inches
0.05	.002	2.8	.110
0.1	.004	3.0	.118
0.15	.006	3.1	.122
0.8	.031	3.5	.138
1.6	.063	5.8	.228
2.0	.079	7.4	.291
2.55	.100		

NOTES:

1. Dimensions are in millimeters.
2. Tolerance is ± 0.13 mm (.005 inch), unless otherwise specified.

FIGURE 1. Configuration and dimensions.

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TABLE I. Electrical characteristics.

AA59532/03-	Ampere rating	Breaking capacity	Voltage drop at rated current. typical mV	Power dissipation at rated current. typical mW	Fusing I^2t at 4 times rated current	
					Typical A^2s	Maximum A^2s
001	63 m A	100 A/125V ac/dc Power factor (p.f.) / cos φ 1	2550	160	0.00011	0.00064
002	100 m A		1770	180	0.0067	0.0016
003	125 m A		1770	220	0.0011	0.0025
004	160 m A		1700	270	0.0018	0.0041
005	250 m A		990	250	0.0058	0.01
006	350 m A		990	350	0.0076	0.02
007	375 m A		990	370	0.013	0.023
008	400 m A		960	380	0.016	0.026
009	500 m A		350	180	0.016	0.04
010	630 m A		290	180	0.020	0.064
011	750 m A		260	200	0.031	0.09
012	1 A		220	220	0.086	0.16
013	1.25 A		220	280	0.14	0.25
014	1.5 A		200	300	0.24	0.36
015	1.6 A		200	320	0.27	0.41
016	2 A		200	400	0.44	0.64
017	2.5 A		190	480	0.97	1.0
018	3 A		190	570	1.3	1.4
019	3.15 A		190	600	1.2	1.6
020	3.5 A		140	490	1.6	2.0
021	4 A		140	560	2.1	2.6
022	5 A		140	700	2.9	4.0
023	6.3 A		110	690	14	32
024	7 A		105	740	16	39
025	8 A		100	800	20	51
026	10 A		80	800	54	96

TABLE II. Opening time characteristics.

Rated current	100 percent of rated current	200 percent of rated current	400 percent of rated current
63mA – 5A	≥ 4 hours	< 1 second	< 10 milliseconds
6.3A – 8A	≥ 4 hours	< 5 seconds	< 50 milliseconds
10A	≥ 4 hours	< 20 seconds	< 60 milliseconds

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Shock resistance. Fuses shall meet the shock requirements in accordance with IEC 60068-2-27, test Ea (981 m/s^2, 6ms).

Solderability (reflow and wave soldering). Fuses shall be able to withstand, without electrical or mechanical damage to the fuse, a wave or reflow solder of +235°C (455°F) for 2 seconds maximum in accordance with IEC 60068-2-58 test Td.

Resistance to soldering heat. Fuses shall meet resistance to soldering heat requirements in accordance with IEC 60068-2-58 test Td (10 seconds at 260°C (500 °F)).

Fuse-link temperature rise ≤ 75 K. PC board track widths shall be ≤ 5 mm (0.196 in.) for fuses rated at 5 A or less, ≥ 5 mm (0.196 in.) for fuses rated at 6.3 and 7 A and ≥ 10 mm (.394 in.) for fuses rated at 8 and 10 A.

Permissible continuous operating current at 23°C (73.5 °F). The permissible continuous operating current at 23°C (73.5 °F) shall be 0.7 times rated current.

Physical specifications.

Materials.

Housing material. The fuse shall be temperature resistance plastic with a UL94 VO flammability rating as a minimum.

Terminal plating. Terminals shall be tin-plated brass. Use of pure tin plating is prohibited as a finial finish and as an undercoat. Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent.

Accommodating fuseholder. Fuses supplied to this CID (5 A rating or less) may be obtained with an accommodating fuseholder under A-A-59532/4 or fuseholders may be obtained separately under CID A-A-59533.

PACKAGING. Packaging shall be as specified in A-A-59532. In addition, fuses may be supplied individually, in a quantity of 750, or in a quantity of 3,000 on a 22.4 mm (.881 inches) wide tape reel in accordance with IEC 60286-3.

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See classification information for PIN format example.

Source of documents.

Commercial Item Description

A-A-59532/4 - Fuse, Inclosed Link, Surface Mount (SM), 125 Volts (V) AC/DC, Quick Acting With a Fuseholder Block.

A-A-59533 - Fuseholder Block, 5 Ampere (A), 125 Volt (V) AC/DC, Surface Mount (SM), General Requirements for.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://dodssp.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Other Publications

Non Government Publications

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60068-2-6 - Environmental Testing – Part 2: Tests - Test Fc: Vibration (Sinusoidal).

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IEC 60068-2-27 - Environmental Testing – Part 2: Tests - Test Ea and Guidance: Shock.

IEC 60068-2-58 - Environmental Testing – Part 2-58: Tests - Test Td - Test Methods for Solderability, Resistance to Dissolution of Metallization and to Soldering Heat of Surface Mounting Devices (SMD).

IEC 60286-3 - Packaging of Components for Automatic Handling - on Continuous Tapes.

(Copies of these documents are available online at http://www.iec.ch/ or from the International Electrotechnical Commission, 3 Rue De Varembe', P.O. Box 131, CH - 1211, Geneve 20, Switzerland.)

UNDERWRITERS LABORATORIES, INCORPORATED (UL)

UL94 - Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances, Standard For.

(Copies of this document is available online at http://www.ul.com/ or from the Underwriters Laboratories, Incorporated, 333 Pfingsten Road, Northbrook, IL 60062-2096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specificatiob sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
61935	Schurter, Incorporated 447 Aviation Boulevard Santa Rosa, CA 95403-1069 Phone number (707) 636-3000 Facsimile number (707) 636-3033 E-mail: mailto:info@schurterinc.com Uniform Resource Locator (URL): http://www.schurterinc.com

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Part number (P/N) supersession data. These CID PINs supersede the following MFR's P/N's as shown. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. P/N supersession data.

CID dash number (see table I) AA59532/03-	MFR's CAGE	MFR's P/N 1/	CID dash number (see table I) AA59532/03-	MFR's CAGE	MFR's P/N 1/
001	61935	3404.0003.11	014	61935	3404.0047.11
002	61935	3404.0004.11	015	61935	3404.0011.11
003	61935	3404.0049.11	016	61935	3404.0012.11
004	61935	3404.0005.11	017	61935	3404.0013.11
005	61935	3404.0006.11	018	61935	3404.0014.11
006	61935	3404.0043.11	019	61935	3404.0048.11
007	61935	3404.0044.11	020	61935	3404.0015.11
008	61935	3404.0007.11	021	61935	3404.0016.11
009	61935	3404.0045.11	022	61935	3404.0017.11
010	61935	3404.0008.11	023	61935	3404.0018.11
011	61935	3404.0046.11	024	61935	3404.0019.11
012	61935	3404.0009.11	025	61935	3404.0020.11
013	61935	3404.0010.11	026	61935	3404.0021.11

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID, the aforementioned commercial products were reviewed and could be replaced by the CID P/N shown. For actual part marking requirements see the marking paragraph of A-A-59532.

MILITARY INTERESTS:

Custodians:
Navy - EC
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing activity:

DLA-CC

Project 5920-2006-011

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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